TEXORA · COURSE NOTES · LECTURE SHEET

Orthographic Projection

Describing solids with flat views — projectors and planes, first-angle versus third-angle projection, view placement and alignment, and the method for drawing the three standard views.

ME-1102 · ENGINEERING DRAWING-I · TOPIC 4 · LEVEL 1 TERM 1

1. The idea: shadows with parallel light

A three-dimensional object must be described on a two-dimensional sheet completely enough to be made from the description. **Orthographic projection** does it with **projectors** — imaginary lines from every point of the object — meeting a **plane of projection** at right angles (that perpendicularity is what “orthographic” means), as if the object cast a shadow in perfectly parallel light from infinitely far away. The resulting view shows the object’s true shape and true dimensions in two axes (those parallel to the plane) with no perspective shrinking. One view cannot show depth, so engineering uses a coordinated set: typically the **front view (elevation)** on the vertical plane (VP), the **top view (plan)** on the horizontal plane (HP), and a **side view** on a profile plane — three flat shadows that between them fix every dimension.

2. The two systems: first angle and third angle

The object sits either before the planes (first angle) or behind them (third angle), and the views are unfolded onto the sheet accordingly:

FEATURE	FIRST-ANGLE PROJECTION	THIRD-ANGLE PROJECTION
Object placed	Between observer and plane (in the first quadrant)	Behind the plane, plane between observer and object (third quadrant)
Top view position	Below the front view	Above the front view
Left-side view position	On the right of the front view	On the left of the front view
Reading rule	Each view is placed on the side opposite to the direction of looking	Each view is placed on the same side as the direction of looking
Used in	ISO / most of Europe and Asia — the convention of this course and	USA and Canada

Because the two systems place views oppositely, every professional drawing carries the **projection symbol** (a truncated cone drawn both ways) in or near the title block, declaring its system; a drawing read in the wrong system is a mirrored part — the most expensive reading error in the subject. Unless told otherwise, this course works in **first angle**.

3. Alignment: the views are one drawing, not three

The views are bound by projection lines, and their positions are not a matter of taste:

- **Front view and top view** share the object's **length** and stand in vertical alignment — every point of the top view lies directly above/below the corresponding point of the front view (45° mitre line or compass transfers carry features across).
- **Front view and side view** share the **height** and stand in horizontal alignment.
- **Top view and side view** share the **width (depth)** — transferred between them by the 45° mitre line drawn in the corner of the layout, or by arcs centred at the layout origin.

These three sharings — length, height, width each appearing twice and required to agree — are the self-checking machinery of the method: if a feature projects correctly through all three views, it is right; the examination sheet is graded largely on this consistency.

4. Drawing the three views: method

1. **Choose the front view:** the direction showing the object's most characteristic shape and most detail (usually its longest dimension horizontal), so that other views carry the least hidden detail.
2. **Lay out the boxes:** mark the overall length × height (front), length × width (top) and width × height (side) rectangles in correct alignment with working space between them, all in light construction lines.
3. **Project feature by feature:** draw each element (face, edge, hole) in the view where it is clearest, then project it into the other two views along the alignment lines — never draw a view alone.
4. **Hidden detail:** edges behind solid material become dashed hidden lines (Topic 2) in the view where they are concealed.
5. **Finish:** darken in the standard line weights, add centre lines to circles and symmetrical features, then dimensions — placed outside the views where possible, each dimension given once only.

> **WORKED EXAMPLE — AN L-BLOCK, READ IN WORDS**
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> Object: an L-shaped block, overall 80 (length) × 40 (width) × 50 (height): a base slab 80 × 40 × 20 with an upright wall 80 × 15 × 30 standing on its back edge. Front view (looking at the 80-side):

the L itself — a rectangle 80 × 50 whose lower 20 is full width and whose upper 30 occupies the back 15 of depth; the step line at height 20 shows across the front region. Top view (first angle, placed below): an 80 × 40 rectangle split lengthwise into two strips — 15 (the wall) and 25 (the open slab). Side view (placed on the left-side direction's opposite side, per first angle): a 40 × 50 rectangle with the step profile: a 20-high portion spanning the 40 width and a 30-high portion over the back 15. Read together, any one view's ambiguity is resolved by the other two — the definition of the method working.

5. Sectional views and conventions (a first meeting)

When interior detail would drown a view in hidden lines, the object is imagined cut by a **cutting plane** and the nearer half removed: a **sectional view**, with cut material hatched (thin lines at 45°) and the cutting plane marked by its chain line with arrows (Topic 2). Standard conventions support all views: threads, springs and repeated features are drawn simplified by rule, and very long parts are shortened with break lines so one sheet can hold them. These conventions are codified (ISO 128 family) — the point here is that orthographic projection is a language with a grammar, and this topic is its first chapter.

6. Glossary — key terms of this topic

TERM	MEANING
Projector	An imaginary line from a point of the object perpendicular to the projection plane.
Front view (elevation)	The view on the vertical plane; the principal view, chosen for maximum information.
Top view (plan)	The view on the horizontal plane, looking down.
First-angle projection	Object between observer and plane; views placed opposite the looking direction (top view below front view).
Third-angle projection	Plane between observer and object; views placed on the looking side (top view above).
Mitre line	The 45° construction line used to transfer depth between top and side views.
Cutting plane	The imaginary plane along which an object is sectioned to reveal its interior.
Hatching	The 45° thin-line pattern marking cut material in a sectional view.

7. Key points and summary

- Orthographic projection describes a solid by coordinated views on perpendicular planes — front, top, side — each true in two dimensions.
- First-angle (course standard): top view below, side views opposite the looking direction; third-angle (North America) reverses the placements; the projection symbol declares which system a drawing uses.
- Views share dimensions pairwise (length, height, width) and are drawn together, feature by feature, with hidden lines dashed and sections hatched.

8. Practice

Written and drawing exercises; brief model answers follow.

1. State, for first-angle projection, where the top view and the right-side view are placed relative to the front view — then give the same for third-angle.
2. Which dimensions are shared by (a) front and top views, (b) front and side views, (c) top and side views, and what construction transfers (c)?
3. A drawing arrives without a projection symbol; its top view is below the front view. In which system was it drawn, and what error follows from reading it in the other?
4. Drawing exercise: draw the three first-angle views (freehand sketch acceptable) of a rectangular block 60 × 40 × 30 with a through-hole of Ø20 drilled vertically through its centre — noting which lines are hidden in which view.
5. Drawing exercise: project the L-block of §4 at scale 1:2 on a sheet layout, using a mitre line for the side view, and verify all three shared dimensions.
6. An object’s front view is a circle and its top view a rectangle. Name the solid and the orientation of its axis.
7. Why is the front view chosen as the most informative direction rather than any other?
8. A view is full of dashed lines describing a complicated interior. What drawing device replaces them, and how is cut material shown?

Answer key

1. First angle: top view below the front view; right-side view (looking from the right) placed on the left of the front view. Third angle: top view above; right-side view on the right.
2. (a) Length; (b) height; (c) width/depth — transferred by a 45° mitre line (or compass arcs about the layout origin).
3. First angle (top view below is its signature). Read as third angle, the part would be made mirrored — top and bottom features interchanged in depth.

4. Self-check: front view 60 × 30 rectangle with two dashed verticals for the hole’s hidden walls; top view 60 × 40 with a full Ø20 circle at centre; side view 40 × 30 with dashed verticals. Length 60 and height 30 must agree across views.
5. Self-check: at 1:2 the overall front rectangle is 40 × 25 mm; the step heights halve to 10 and 15 mm; mitre-transferred depths must match the top view’s 15/25 split (7.5/12.5 mm at scale).
6. A cylinder with its axis perpendicular to the VP — horizontal, pointing at the viewer along the line of sight: the end face gives the circle, the curved surface’s silhouette the rectangle. (A cone so placed would give a triangle, not a rectangle, in the top view.)
7. The front view anchors the alignment of all other views and carries the dimensions the others share; choosing the most descriptive direction minimises hidden lines and ambiguity everywhere else.
8. A sectional view: the object is cut by a marked cutting plane, the front portion removed, and cut material hatched with 45° thin lines — interiors appear as visible outlines instead of dashes.
